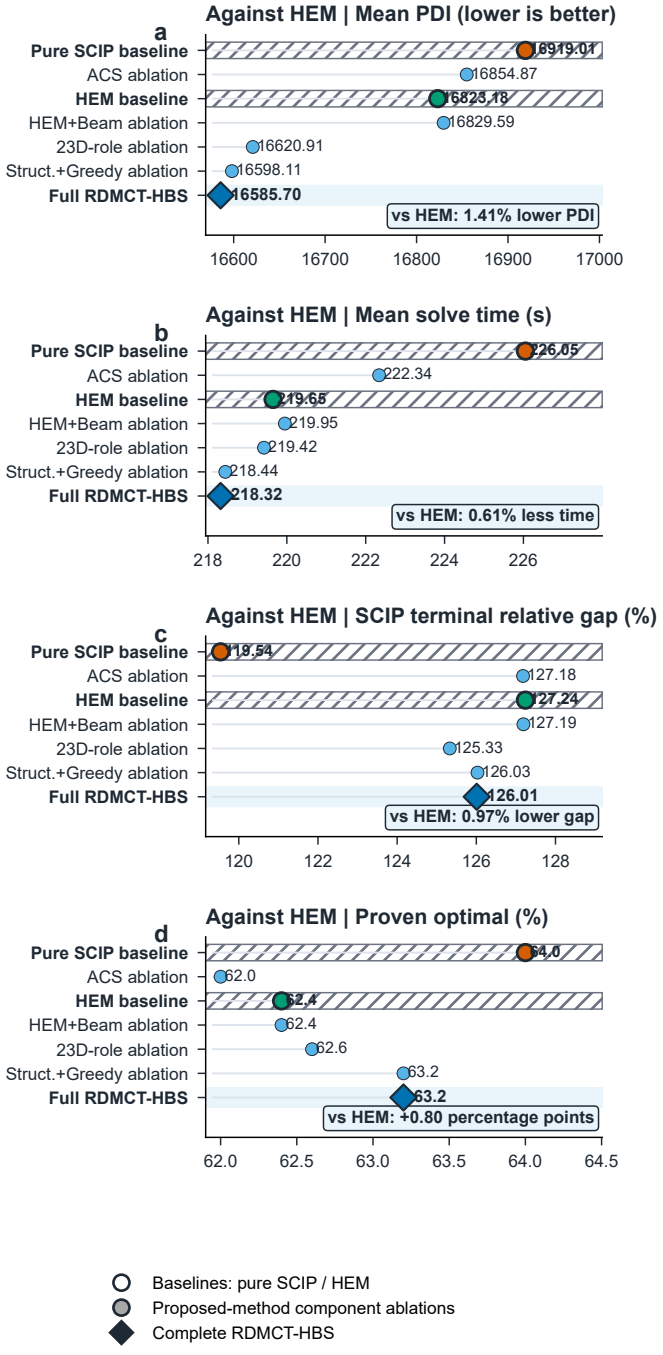

a
Against HEM | Mean PDI (lower is better)
Pure SCIP baseline
ACS ablation
HEM baseline
HEM+Beam ablation
23D-role ablation
Struct.+Greedy ablation
Full RDMCT-HBS
16919.01
16854.87
16823.18
16829.59
16620.91
16598.11
16585.70
vs HEM: 1.41% lower PDI
16600
16700
16800
16900
17000
b
Against HEM | Mean solve time (s)
Pure SCIP baseline
ACS ablation
HEM baseline
HEM+Beam ablation
23D-role ablation
Struct.+Greedy ablation
Full RDMCT-HBS
226.05
222.34
219.65
219.95
219.42
218.44
218.32
vs HEM: 0.61% less time
218
220
222
224
226
c
Against HEM | SCIP terminal relative gap (%)
Pure SCIP baseline
ACS ablation
HEM baseline
HEM+Beam ablation
23D-role ablation
Struct.+Greedy ablation
Full RDMCT-HBS
127.18
127.24
127.19
125.33
126.03
126.01
vs HEM: 0.97% lower gap
120
122
124
126
128
d
Against HEM | Proven optimal (%)
Pure SCIP baseline
ACS ablation
HEM baseline
HEM+Beam ablation
23D-role ablation
Struct.+Greedy ablation
Full RDMCT-HBS
62.0
62.4
62.4
62.6
63.2
63.2
vs HEM: +0.80 percentage points
62.0
62.5
63.0
63.5
64.0
64.5
Baselines: pure SCIP / HEM
Proposed-method component ablations
Complete RDMCT-HBS